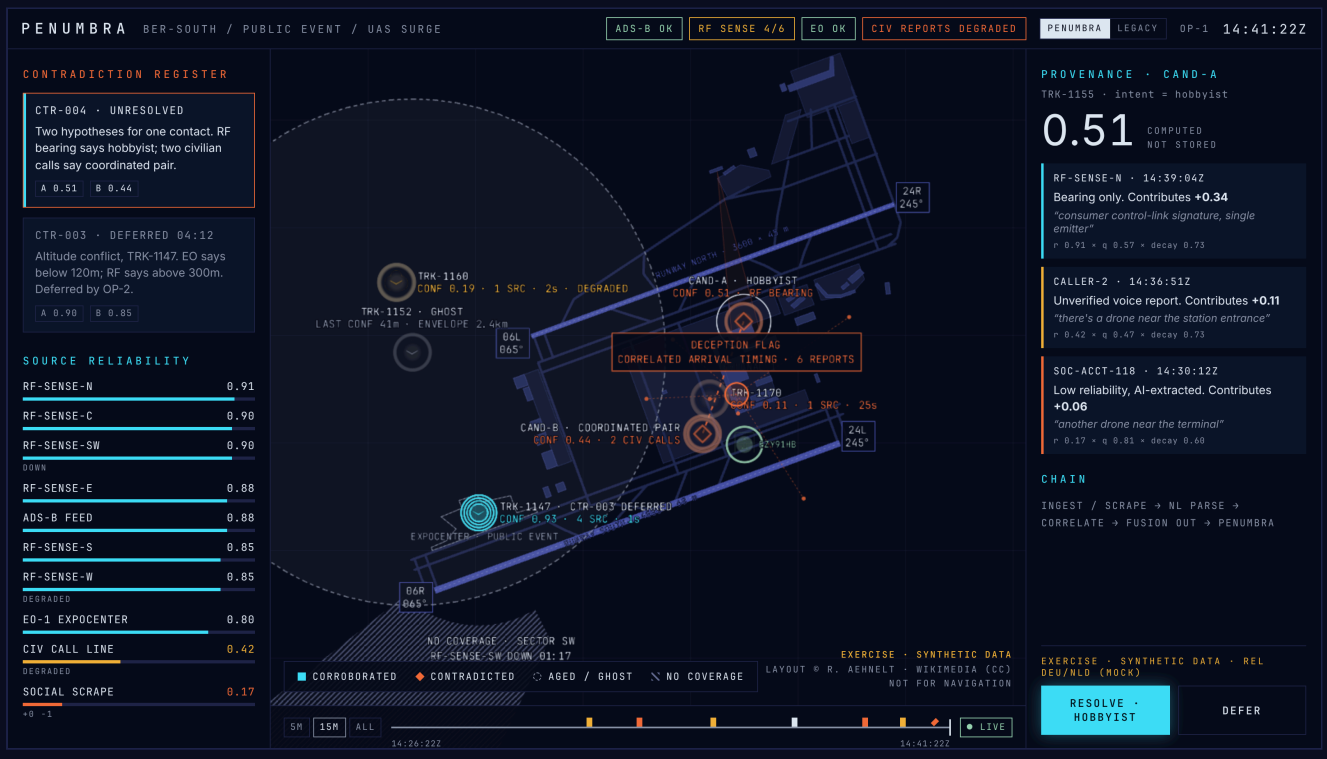
PENUMBRA

Threat interface, Phase 0 prototype. How to run it, read it, decide with it, and show it.



STRATOSPHERE SOLUTIONS
BERLIN · NOVEMBER 2026

VERSION 0.5 · MILESTONES M1–M5
EXERCISE · SYNTHETIC DATA

// ABOUT THIS MANUAL

WHAT PENUMBRA IS

PENUMBRA is a confidence and decision-support layer for the common operational picture. It sits above sensor fusion and draws what fielded pictures leave out: how sure we are, how old the information is, where it came from, where sources disagree, and where we cannot see at all.

01

Confidence on the picture

Every contact shows how much evidence stands behind it, and every number breaks down into the sources that make it up.

02

Disagreement kept

When sources disagree, both readings stay on screen until an operator decides. Nothing is averaged away.

03

Blindness made visible

"Nothing there" and "we cannot see there" look different. Stale contacts age visibly instead of looking current.

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PHASE 0 PROTOTYPE

All data is synthetic: callsigns, sensors, callers and accounts are invented. The scenario is a drone surge at Berlin Brandenburg (EDDB) during a public event at the ExpoCenter. The system is not accredited and must not be used operationally.

// 01 · GETTING STARTED

INSTALL AND RUN

PENUMBRA runs on a normal laptop. It needs no internet connection once installed, no map tiles and no cloud service.

YOU NEED

- **Node.js 22 or newer** (the current LTS from nodejs.org) and **pnpm 10** (`npm install -g pnpm`).
- A copy of the repository, on the branch with the prototype.
- A recent Chrome, Edge or Firefox. A laptop with a graphics chip renders smoothly; a virtual machine without one is slow.

FIRST RUN

From the `penumbra` folder, run each line separately:

```
pnpm install          # once, and after every update
pnpm dev              # feed server + client
```

Open **http://127.0.0.1:5173**. Playback starts at 14:38:22; the moment from the pitch deck (slide 7) arrives at 14:41:22, three minutes in. After about 40 seconds the RF reading arrives and contradiction CTR-004 appears.

LIVE

pnpm dev

Feed server and client together. The server is the single source of truth: every client sees the same picture, decisions go into its append-only log, and session logs are stored.

OFFLINE

pnpm client

The client alone. With no server it plays the scenario in the browser. Everything works, including decisions; session logs are kept for download. Good for travelling demos.

WINDOWS POWERSHELL

Windows PowerShell 5.1 does not accept `&&`: run commands one per line. If PowerShell refuses to run `npm` or `pnpm` ("running scripts is disabled"), run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once, or use `npm.cmd` and `pnpm.cmd` instead.

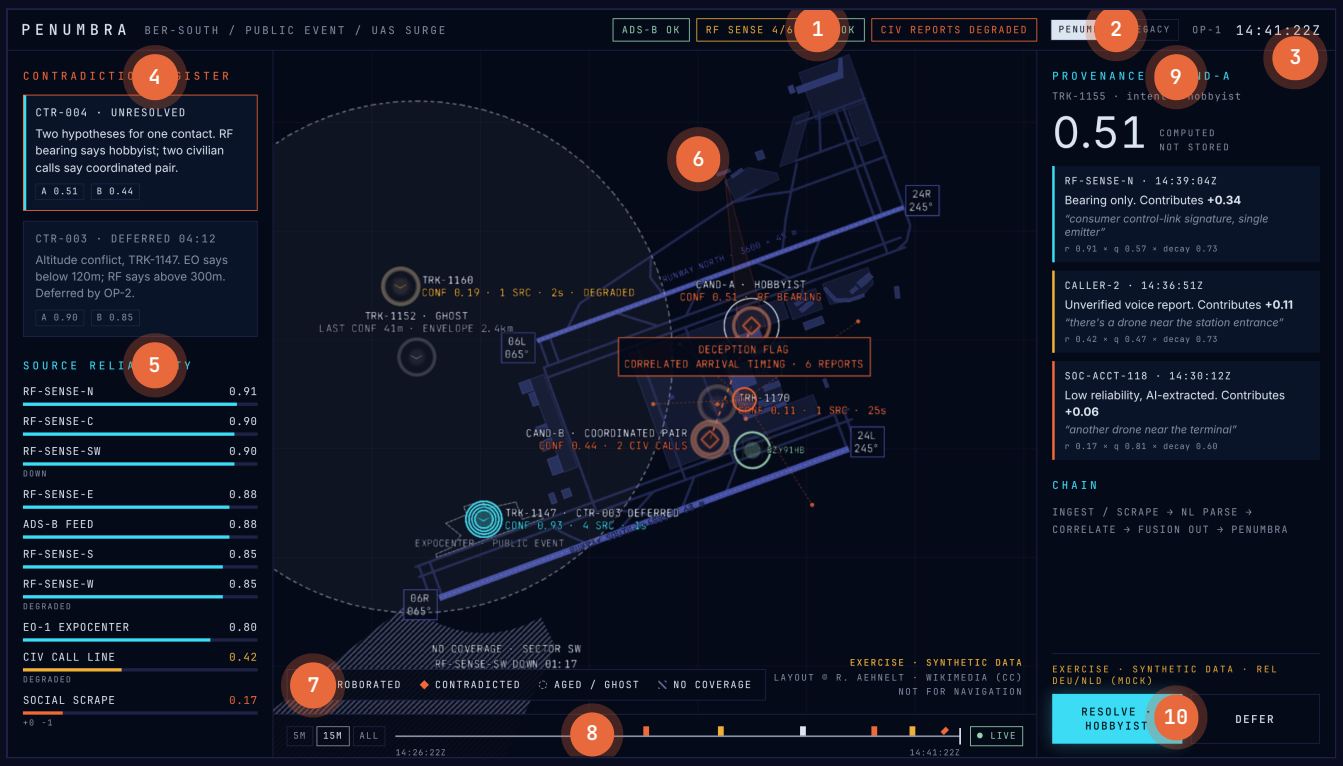
STOPPING AND RESTARTING

`Ctrl` + `C` in the terminal stops everything. The ⟲ button in the header restarts the scenario from 14:38:22; `Space` pauses and resumes; the speed menu runs it at up to 16×.

// 02 · THE SCREEN AT A GLANCE

ONE SCREEN, FIVE AREAS

The picture in the middle, what needs a decision on the left, the evidence behind the selected item on the right, time along the bottom.

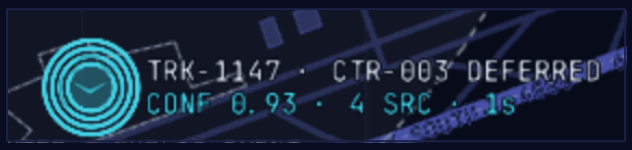


KEY FRAME 14:41:22Z · THE COMPOSITION OF PITCH-DECK SLIDE 7

1. **Feed health.** Each sensor family at a glance: ADS-B OK, RF SENSE 4/6, CIV REPORTS DEGRADED.
2. **View switch.** PENUMBRA or LEGACY, the same data drawn the fielded way (V).
3. **Operator and clock.** Your operator id, and the time of the picture (amber when replaying).
4. **Contradiction register.** Open disagreements, newest first, each explained in a sentence.
5. **Source reliability.** How trustworthy each source has proven, updated by confirmed and false reports.
6. **The picture.** Contacts, uncertainty, coverage and flags on the EDDB layout. Click a contact to select it.
7. **Legend.** Corroborated, contradicted, aged or ghost, no coverage.
8. **Timeline.** Drag to see the picture as it stood at any past moment. Marks jump to key events.
9. **Provenance.** The selected item's confidence and every source that makes it up.
10. **Decisions.** Resolve or defer the selected contradiction (R , D).

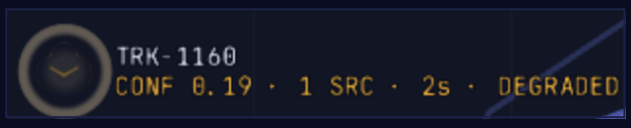
CONFIDENCE YOU CAN SEE

A fielded picture draws a confirmed track and an unverified report with the same symbol. PENUMBRA draws how much evidence stands behind each contact.



Strong: four rings, crisp, full colour

TRK-1147 over the ExpoCenter: four independent sensors agree, confidence 0.93.



Weak: one ring, blurred, washed out

TRK-1160: one degraded sensor, confidence 0.19. Same symbol type, very different picture.

THREE VISUAL PROPERTIES

PROPERTY	MEANS	HOW TO READ IT
Ring count	Independent sources	One ring per independent source, up to four. Many reports from one sensor still count as one.
Edge sharpness	Confidence	Crisp edges mean high confidence; edges blur as confidence falls.
Colour strength	Confidence	Full colour when confident, fading towards grey when not. The symbol fades too.

COLOURS

COLOUR	MEANING
Cyan	Corroborated: at least two independent sources and confidence of at least 0.6.
Orange	An unresolved contradiction, or a deception flag. Needs attention.
Amber	The strongest source behind the contact is degraded.
Grey	A ghost (no recent fix), or a single weak source.
Green	Civil aircraft reporting by transponder (ADS-B). Context, labelled small.

WHERE THE NUMBER COMES FROM

Each report counts by **source reliability × report quality × age decay**. Each independent source gets one vote: its strongest report. The votes combine like independent chances, so agreeing sources compound and one weak report stays weak. The provenance panel shows each source's share, and the shares always add up to the number on screen.

AGE, DISAGREEMENT, BLINDNESS



Ghosts: old information ages

When fixes stop (after 30 s for a small drone), the contact becomes a **ghost**: a dashed symbol inside a dashed area showing where it is likely to be now. The area grows with time. A faint outer ring marks the furthest it could physically have flown. Ghosts stay for an hour, labelled with the age of the last fix.



Two readings, neither averaged away

When sources disagree about a contact whose position is uncertain, each reading is drawn as a **hollow orange diamond** where its evidence puts it, linked by a dashed line: CAND-A (hobbyist, from an RF bearing) and CAND-B (coordinated pair, from two callers). A **wedge** marks a bearing-only report: the sensor knows the direction, not the distance. A well-tracked contact with an attribute conflict keeps one symbol and a tag, e.g. "TRK-1147 · CTR-003 DEFERRED".



Coverage: "cannot see" is not "nothing there"

Plain ground	Covered by a working drone sensor. Empty means empty.
Dense hatch	A sensor that covered it is down, e.g. "NO COVERAGE · SECTOR SW".
Faint hatch	No drone sensor has ever covered it.
Amber tint	Only degraded sensors see here.



Deception flag

Six or more low-reliability accounts reporting the same thing within 60 seconds, in one area, is not how independent witnesses behave. PENUMBRA raises a **deception flag** with lines to each report and **counts the burst as one vote, not six**: here the planted "swarm" scores 0.11 instead of 0.51. Nothing is deleted or marked false. The flag clears ten minutes after the last report.

REGISTER, SOURCES, PROVENANCE

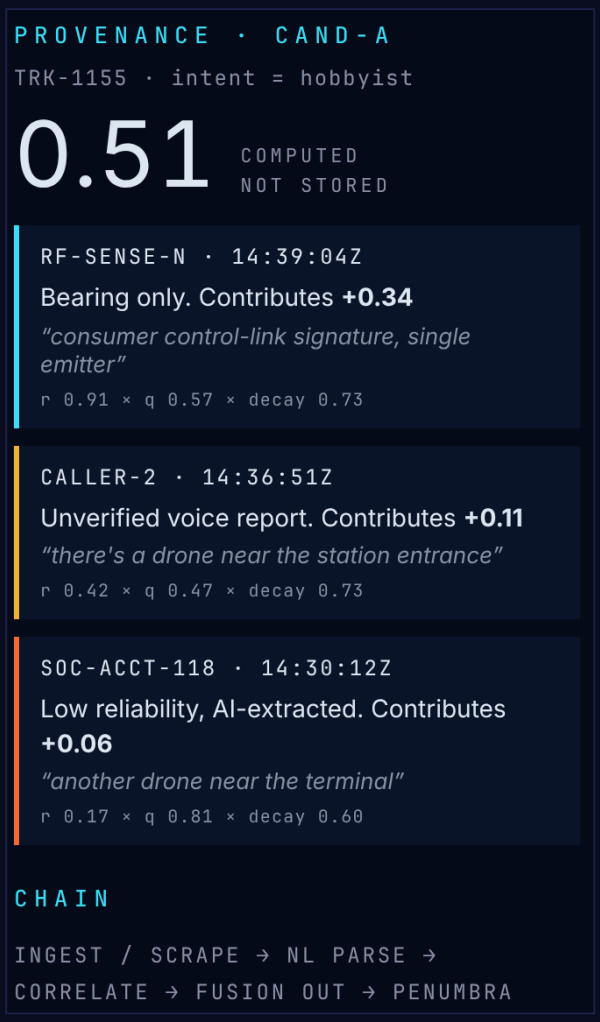
CONTRADICTION REGISTER

Every disagreement between sources, newest first, written as a sentence: *"Two hypotheses for one contact. RF bearing says hobbyist; two civilian calls say coordinated pair."* Click a card to select its leading candidate.

STATUS	MEANS
UNRESOLVED	Both readings live, no decision yet.
DEFERRED 04:12	An operator parked it; the timer counts up.
RESOLVED · HOBBYIST	Decided; the rejected reading is struck through.
REOPENED	Fresh evidence for a rejected reading arrived after the decision: decide again.
LAPSED	Support for all but one reading has decayed.

SOURCE RELIABILITY

How trustworthy each source has proven. It starts from a prior, moves up when police confirm a report and down when one is shown false. Bars are cyan from 0.75, amber from 0.40, orange below. Notes show DOWN, DEGRADED and the verdict count (`+1 -0`). After the first resolution, **OPERATOR** appears too: decisions are evidence, so they carry a reliability.



PROVENANCE

The big number is the selected item's confidence, **computed, not stored**: recalculated from the evidence every time you look. Below it, one card per source:

- **Who and when:** `RF-SENSE-N · 14:39:04Z`
- **What kind of report** in plain words, and its share: "Bearing only. Contributes +0.34"
- **The original text**, for human reports.
- **The factors:** reliability × quality × decay.

Chain shows the processing path, e.g. INGEST / SCRAPE → NL PARSE → CORRELATE → FUSION OUT → PENUMBRA. The releasability marking sits above the buttons.

RESOLVE AND DEFER

Select a candidate, on the map or through its register card, and decide. Decisions go into the append-only log with your operator id and the time; nothing can be edited afterwards.

RESOLVE · R

Accept one reading

- Your decision becomes evidence for the accepted reading, from the source **OPERATOR** (reliability 0.80, fading over about half an hour). It appears as its own card in provenance.
- Earlier reports for the rejected reading still say something is there, so they now count for the accepted one.
- The rejected reading drops to zero.

DEFER · D

Park it, knowingly

- No number changes. The status becomes DEFERRED with a running timer.
- The contact keeps its own colour and carries a tag (e.g. "CTR-003 DEFERRED"), so it no longer competes for attention.
- A deferred contradiction can still be resolved later; it cannot be deferred twice.

WHEN A DECISION IS REOPENED

If fresh, live evidence for a rejected reading arrives after your decision, the contradiction reopens and returns to the register: the system does not quietly overrule new evidence. In the scenario, resolve CTR-004 before 14:39:10 and the third caller ("two of them, one following the other") reopens it.

WHEN A BUTTON IS GREYED OUT

SITUATION	WHY
Whole contact selected, not a candidate	Resolve needs a reading: select CAND-A or CAND-B.
No open contradiction on the contact	Nothing to decide.
Replay mode	Decisions are taken at the live picture only. Press `End` to return.
Already resolved, already deferred	The server and the offline player refuse it with the same message.

LIVE AND OFFLINE BEHAVE THE SAME

In live mode the decision is sent to the feed server, written to its log and shown to every connected client; you get a confirmation such as "CTR-004 resolved as hobbyist · OP-1". Offline, the same rules apply locally. Set your operator id with `?op=OP-3`.

// 07 · LOOKING BACK

REPLAY: EXACTLY WHAT WAS ON SCREEN

Drag the timeline to any past moment and the picture is rebuilt exactly as it stood then, including the decisions taken up to that moment and none after.



REPLAY OF 14:36:00 SEEN FROM 14:41:22 · CTR-003 STILL UNRESOLVED, RF 5/6, CTR-004 NOT YET OPEN

Using the timeline

- **Drag** anywhere on the track. The right edge is live.
- **Marks** jump to key moments: contradictions, **decisions**, sensor changes, verdicts, and deception flags (diamond).
- **5M · 15M · ALL** sets the window. Close marks stack in two rows.
- ← → step 5 s; with `Shift`, 1 min. `End` or `Esc` returns to live.

You cannot mistake it for live

- Amber banner: "REPLAY 14:36:00Z · 05:22 behind live".
- Amber clock tagged REPLAY, amber timeline.
- Resolve and defer are disabled.
- Playback continues underneath; "LIVE" brings you back to the current picture.

Use it in after-action review: what could the operator see at the moment they decided?

THE SAME DATA, THE FIELDED WAY

Press V (or LEGACY in the header) to redraw the same moment by the rules fielded systems follow. It is a fair copy, built as a baseline for comparison, not a strawman.



LEGACY VIEW AT 14:41:22 · SAME SNAPSHOT AS THE COVER

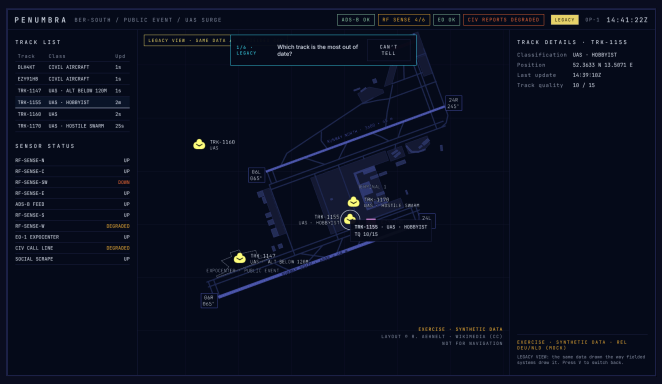
FIELDED BEHAVIOUR	WHAT YOU SEE IN LEGACY
One fused track per contact	CAND-A and CAND-B become one TRK-1155, labelled only "HOBBYIST".
Fusion picks a best estimate	The competing reading disappears; there is no contradiction register.
Uniform standard symbols	TRK-1160 (0.19, one degraded sensor) looks identical to TRK-1147 (0.93, four sensors).
No ageing, then a drop timer	Tracks look current until dropped after 5 minutes; the 41-minute ghost is gone.
No coverage, no adjudication	No blind sector; the planted posts appear as an ordinary "UAS · HOSTILE SWARM" track.
Track quality as metadata	Available as a number (e.g. 14 / 15) in track details, when you click a track. Not on the map.

The panels change to what fielded systems show: a track list with update times, sensor status, and track details. Nothing a fielded system shows is hidden; nothing it lacks is added.

// 09 · LEGIBILITY PROBE

MEASURE WHO READS IT BETTER

A six-question test that measures how quickly and correctly people read the picture in each view. The results table is the evidence for the DIANA submission.



A TRIAL IN THE LEGACY VIEW

The three questions, asked in both views

- Which contact is the least reliable?
- Where can we not see drones? (click a point)
- Which track is the most out of date?

Answers are clicks on the map, timed from the moment the picture is painted. "Can't tell" counts as wrong.

Running a session

1. Start `pnpm dev` so answers are stored on the server.
2. Open **http://127.0.0.1:5173/?probe**.
3. Enter the participant id and press **Start**.
4. For each trial: read the question, press **Show the picture**, click the answer.
5. The last screen shows the participant's result per view.
6. After all sessions: `pnpm probe:report`.

Why it is fair

- Each person sees the two views at **different moments**, so the second view gains nothing from the first.
- Which view comes first alternates between participant ids.
- Right answers are **computed from the system's own picture** for the view shown, never written by hand.

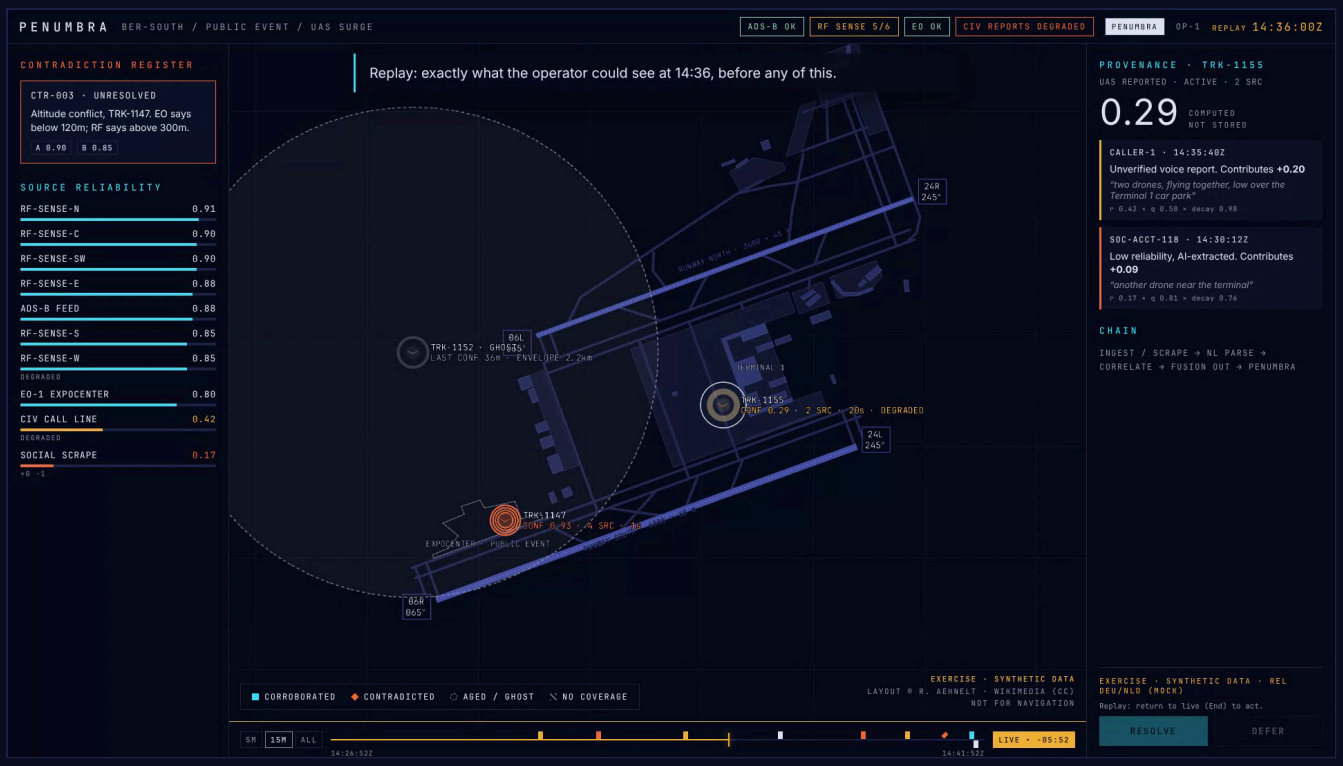
```
pnpm probe:report            # ./sessions
pnpm probe:report -- a.jsonl # downloads
```

Prints a table per view and per question: answers, % correct, median time.

> BEFORE REAL PARTICIPANTS
>
> Choose the groups (the plan names reservists, civil protection, air traffic and emergency operations), prepare a one-paragraph consent note, and pilot with two or three colleagues to check the questions read clearly. Automated test runs are robot clicks, not evidence.

SHOW IT WITHOUT TOUCHING IT

A captioned walkthrough that plays itself: for stands, briefings and the supplementary video.



THE DEMO'S CLOSING REPLAY, CAPTURED AT 1920 × 1080

Run it

```
pnpm demo            # opens the browser
pnpm demo:record     # saves a video
```

The demo restarts the scenario and plays the scripted ending: police confirm the hobbyist drone, OP-1 resolves CTR-004. Keys, transport and the view switch are locked so a stray key cannot derail it. Add `?capture=1080p` to render at exactly 1920 × 1080 for a screen recorder such as OBS.

Record the real video on a laptop with a graphics chip

`pnpm demo:record` records in real time to `recordings/`. Without a graphics chip (cloud machines, some virtual machines) the frame rate is too low for a usable video.

The walkthrough, about 4 minutes

14:38:23	The event, the surge of reports
14:38:40	TRK-1147: four sensors, four rings
14:39:06	CAND-A vs CAND-B, neither averaged away
14:39:40	TRK-1152, last seen 41 minutes ago
14:40:07	A sensor fails; sector SW goes blind
14:41:02	Six accounts, counted as one vote
14:41:12	0.51 = 0.34 + 0.11 + 0.06
14:41:24	Seven seconds of LEGACY
14:41:34	Police confirm; OP-1 resolves
after end	Replay of 14:36, closing line

Captions and timing live in `apps/client/src/demo/director.ts`; changing a caption is a one-line text edit.

KEYS, URL OPTIONS, COMMANDS

KEYBOARD

Space	Play / pause
R	Resolve the selected candidate
D	Defer the selected contradiction
V	Switch PENUMBRA / LEGACY
← →	Replay 5 s back / forward
Shift + ← →	Replay 1 min back / forward
End Esc	Back to live

COMMANDS (IN `PENUMBRA/`)

`pnpm dev`	Feed server and client
`pnpm client`	Client only (offline fallback)
`pnpm feed`	Feed server only (`--speed`, `--manual`, `--paused`)
`pnpm demo`	Captioned walkthrough
`pnpm demo:record`	Record the walkthrough
`pnpm probe:report`	Legibility probe results
`pnpm frame`	The key frame as text
`pnpm shot`	Screenshot to `docs/screens/`
`pnpm check`	Type check and all tests
`pnpm smoke`	Live end-to-end browser test
`pnpm build`	Static client (runs offline)
`pnpm manual`	Rebuild this manual

URL OPTIONS

Add to the address, e.g. `127.0.0.1:5173/?view=legacy&speed=4`

`?t=14:41:22`	Offline, frozen at a moment
`?mode=offline`	Play in the browser, no server
`?speed=4`	Playback speed (up to 16)
`?op=OP-3`	Operator id for decisions
`?view=legacy`	Open in the LEGACY view
`?replay=14:36:00`	Open replaying a moment
`?scripted`	Offline: keep the scripted decisions
`?probe`	Legibility probe
`?demo`	Captioned walkthrough
`?capture=1080p`	Exactly 1920 × 1080, for recording
`?stress=500&stats`	500 extra drones and an fps meter
`?ws=…`	Feed address (default: same machine)

> PERFORMANCE CHECK
>
> Open `?stress=500&stats&speed=4` on the demo laptop and read the fps meter while it plays. The target is 60 fps with 500 contacts.

WHEN SOMETHING GOES WRONG

SYMPTOM	FIX
"The token '&&' is not a valid statement separator"	Windows PowerShell 5.1: run the commands one per line.
"npm / pnpm is not recognized"	Install Node.js LTS from nodejs.org, open a new terminal, then `npm install -g pnpm`.
"running scripts is disabled on this system"	`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or use `npm.cmd` / `pnpm.cmd`.
Port 5173 or 8787 already in use	Close the other program, or a previous PENUMBRA still running (`Ctrl` + `C` in its terminal).
Header says OFFLINE, not LIVE FEED	The feed server is not running: use `pnpm dev`, not `pnpm client`.
Map is slow or jerky	No graphics acceleration (virtual machine, remote desktop). Use a laptop with a graphics chip.
Resolve does nothing	Select a candidate (CAND-A / CAND-B), and make sure you are not in replay.

FILES AND LOGS

`sessions/*.jsonl`	Every operator interaction, one file per browser session
`/log.jsonl`	The server's append-only picture log (address of the feed server)
`recordings/`	Demo videos
`docs/`	Design notes per milestone, screenshots, this manual
`scenarios/*.yaml`	The scenario, editable without code

Together, the session log and the picture log let you replay exactly what an operator saw next to what they did and when.

GLOSSARY

Confidence	How much evidence stands behind a contact or reading, 0 to 1.
Origin	An independent source: one sensor, one caller, one account.
Hypothesis	One reading of a contact (e.g. intent = hobbyist), labelled CAND-A, CAND-B.
Contradiction	Two readings with live support at once (CTR-nnn).
Ghost	A contact without a recent fix, drawn as a growing likely area.
Provenance	The sources behind a number, with their shares and processing chain.
Track quality	LEGACY's 0–15 number, shown only in track details.

WHAT THIS PROTOTYPE IS, AND IS NOT

DATA

Synthetic, end to end

The scenario, sensors, callsigns, callers and social accounts are invented. Correlation of reports to contacts is scripted. Use the screens to explain the concept, not to describe a real event.

MAP

EDDB layout, traced

The airport layout is traced from "Airport Layout 08/2026" by R. Aehnelt via Wikimedia Commons (Creative Commons), credited on screen. **Confirm the exact licence before any public use**: a share-alike licence would apply to the traced map data. Not for navigation.

STATUS

Phase 0, not accredited

A prototype for the NATO DIANA 2028 challenge call. No accreditation, no real sensor feeds, no access control. Do not use operationally or with real data.

OPEN

Known limits

60 fps with 500 contacts is not yet measured on target hardware. Replaying very long sessions is slow. Some timeline marks overlap in the widest window. The probe has been tested by automation only, not with participants.

DESIGN DECISIONS TAKEN IN REVIEW

- **A resolution is evidence.** The operator's decision counts towards the accepted reading and can be outweighed by fresh evidence (reopen).
- **A flagged burst is one vote.** The system never deletes or overrules reports; it refuses to count coordinated ones as independent.
- **Ghosts show the likely area,** with the physical maximum as a faint outer ring.